

Table 1: Discrete Distributions

Distribution	Known Abbreviation	Pmf	Expected Value	Variance
Bernoulli	$X \sim Bern(p)$	$p_X(k) = p^k(1-p)^{1-k}$ where $k = 0, 1$	$\mu = p$	$\sigma^2 = p(1-p)$
Binomial	$X \sim Bin(n, p)$	$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$ where $k = 0, 1, \dots, n$	$\mu = np$	$\sigma^2 = np(1-p)$
Geometric	$X \sim Geom(p)$	$p_X(k) = p(1-p)^{k-1}$ where $k = 1, 2, \dots$	$\mu = \frac{1}{p}$	$\sigma^2 = \frac{1-p}{p^2}$
Negative Binomial	$X \sim NBin(r, p)$	$p_X(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$ where $k = r, r+1, \dots$	$\mu = \frac{r}{p}$	$\sigma^2 = \frac{r(1-p)}{p^2}$
Poisson	$X \sim Pois(\lambda)$	$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$ where $k = 0, 1, 2, \dots$	$\mu = \lambda$	$\sigma^2 = \lambda$

Table 2: Continuous Distributions

Distribution	Known Abbreviation	Pdf	Expected Value	Variance
Normal distribution	$X \sim N(\mu, \sigma^2)$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty$	μ	σ^2
Exponential distribution	$X \sim Exp(\lambda)$	$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$